Date:

COAL ASSIGNMENT # 02

QUESTION 01:

Name: Izaan Aamir
Roll No: 25K-0059
Section: BAI-3A

```
.code
main PROC
    mov ax, 3

    movzx eax, ax
    movzx ebx, ax
    movzx ecx, ax

    shl eax, 4
    shl ebx, 3
    shl ecx, 1

    add eax, ebx
    add eax, ecx

    call WriteDec
    call Crlf

    exit
main ENDP
END main
```

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QUESTION 02 and 03 (This code also solves Q3 since 16 x 2 = 32 bits)

```
.data
    val DWORD 02345ABCDh

.code
main PROC
    mov eax, val
    mov edx, 0

    shld edx, eax, 32
    shl eax, 32
    call DumpRegs

    mov eax, val
    mov edx, 0

    shrd edx, eax, 32
    shr eax, 32
    call DumpRegs

exit
in ENDP
D main
```

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QUESTION 04:

```
.data
    BSeconds BYTE ?
    BMinutes BYTE ?

.code
    mov dx , 0101100111011010b

    mov ax, dx
    and ax, 0000000000011111b
    mov BSeconds, al

    mov ax , dx
    shr ax, 5
    and ax, 0000000000111111b
    mov BMinutes, al

    call DumpRegs

exit
main ENDP
END main
```

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Question 05:

1. add ax, 90EAh

7F80h + 90EAh = 1106Ah

AX = 106Ah , CF = 1

2. adc dx, 1

9A3Fh + 1 + 1(CF) = 9A41 h

DX = 9A41h , CF = 0

3. stc

CF = 1

4. add ax, 8120h

106Ah + 8120h = 918Ah

Ax = 918Ah , CF = 0

5. adc dx, 0FFFFh

9A41h + 0FFFFh = 19A40h

DX = 9A40h , CF = 1

Final Contents:

AX = 918Ah

DX = 9A40h

CF = 1

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QUESTION 06:

a) test ax, ax

b) test bx, 1

c) test dx, dx

d) test dx, 8000h

e) test bl, bl

QUESTION 07:

10h + 5h + 1h = 16h

Dx = 16h

QUESTION 08:

- Line 8, 9:
  - we need to write OFFSET to store addresses in esi and edi

- Line 15, 16:
  - we need to use INC instead of DEC to move forward with the subtraction.

- Line 13:
  - sbb written instead of sub

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QUESTION 09:

```
.data
    marks BYTE ?
    msg1 BYTE "Enter your marks: ", 0
    msg2 BYTE "You achieved an A grade!", 0
    msg3 BYTE "You achieved a B grade!", 0
    msg4 BYTE "You achieved a C grade!", 0
    msg5 BYTE "You achieved an F grade!", 0
    msg6 BYTE "Invalid marks entered!", 0

.code
main PROC
    mov edx, OFFSET msg1
    call WriteString
    call ReadInt
    call Crlf

    cmp eax, 100
    ja invalidInput

    cmp eax, 80
    jae gradeA

    cmp eax, 60
    jae gradeB
```


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```
    cmp eax, 40
    jae gradeC

    jmp failed

    gradeA:
        mov edx, OFFSET msg2
        jmp done
    gradeB:
        mov edx, OFFSET msg3
        jmp done
    gradeC:
        mov edx, OFFSET msg4
        jmp done
    failed:
        mov edx, OFFSET msg5
        jmp done
    invalidInput:
        mov edx, OFFSET msg6
        jmp done

done:
    call WriteString
    call Crlf

    exit
main ENDP
END main
```


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QUESTION 10:

EAX = 33

EBX = 10

QUESTION 11:

```
.code
main PROC
    mov eax, num1
    cmp eax, num2
    jle elif

    mov eax, num3
    cmp eax, num4
    jz elif

    add eax, num1
    mov result, eax
    jmp done

elif:
    mov eax, num2
    cmp eax, num4
    jnz finalElse

    sub eax, num1
```


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```
        mov result, eax
        jmp done

    finalElse:
        mov result, 0
    done:
        exit
main ENDP
END main
```

QUESTION 12:

```
.code
main PROC

_While1:
    mov eax, op1
    cmp eax, op2
    jg _endWhile1

_While2:
    mov eax, op1
    cmp eax, op2
    jg _endWhile2

    cmp eax, x
    jl else1.
```


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```
cmp eax, y
jg else1

add z, 10
jmp done

else 1:
    sub z, 10
    jmp done
done:
    dec op1
    jmp _While2

_endWhile2:
    jmp _While1

_endWhile1:
    exit

main ENDP
END main
```


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